Disjoint sets: merge groups, never split them

The partition changes as groups merge. The query asks: **are these two elements in the same group now?**

Question	Method
How do we identify a set?	A list's set record, or a forest root
How much must a merge change?	Move the shorter list; attach the lower-rank root
Why can later finds get cheaper?	Path compression and amortized sequence analysis

Prerequisites: pointers, rooted trees, asymptotic notation, and the potential method. Rank, level, and iter are defined here.

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Three operations maintain one partition

Every created element belongs to exactly one nonempty set; distinct sets are disjoint.

Operation	Precondition	Result
MAKE-SET(x)	x has not been created	Create singleton {x}
FIND(x)	x exists	Return its set's representative
UNION(x,y)	Both elements exist	Merge their sets; do nothing if already together

A representative identifies a set; it need not be its smallest key. Finds on one set return the same representative; different sets have different representatives.

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Count calls separately from successful merges

Create n singletons first. Let m count external MAKE-SET, FIND, and UNION calls, so $m \geq n$. Do not count UNION's internal FIND calls again in m.

Quantity	Contract
Successful merges	At most n−1, because each removes one set
UNION calls	Repeated calls are allowed; no n−1 bound
LINK(r,s)	Accepts two **distinct roots**, not arbitrary elements

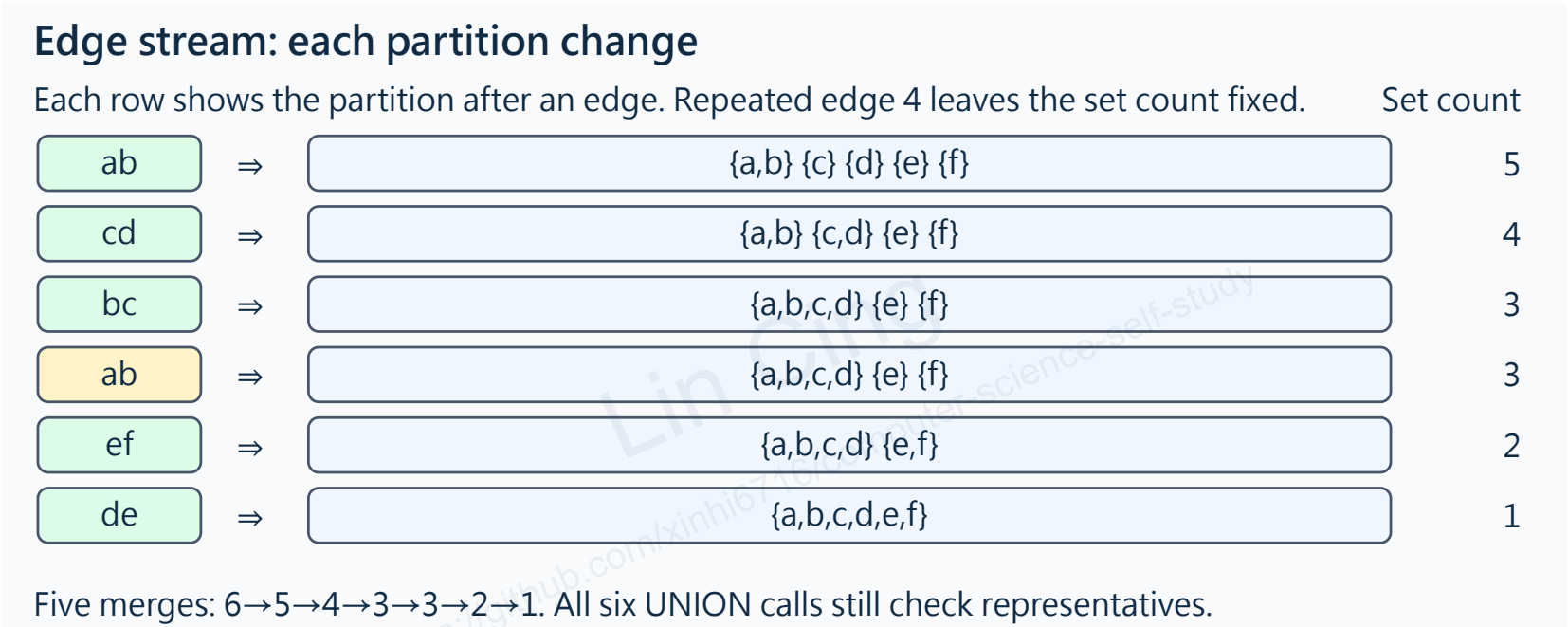
> UNION finds both roots and returns immediately if they agree. **Never LINK a root to itself or increase its rank for a repeated merge.**

FIND may change parents while preserving the partition. Deletion, splitting, and rollback require additional structures and analysis.

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Undirected connectivity: what does a new edge merge?

Create a set for each vertex, then UNION each edge. The diagram uses ab, cd, bc, ab, ef, de.



The fourth edge ab is repeated: the call counts, but the partition does not change. Six UNION calls cause five successful merges.

Why are these exactly the connected components?

Invariant: the components of the processed-edge subgraph equal the maintained partition.

Step	Connectivity	Partition
No edges yet	Each vertex is isolated	One singleton per vertex
New edge within a component	Already connected	UNION changes nothing
New edge between components	Joins exactly those two components	UNION merges exactly their two sets

Induction gives FIND(u)=FIND(v) exactly when an undirected u–v path exists.

DFS/BFS finds static components in $\Theta(n+|E|)$. DSU also supports **connectivity queries as edges arrive**. Ignoring edge directions yields weak components, not SCCs.

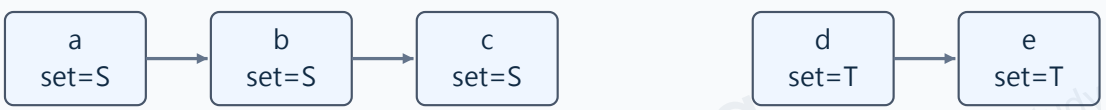
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Lists: joining tails does not finish the merge

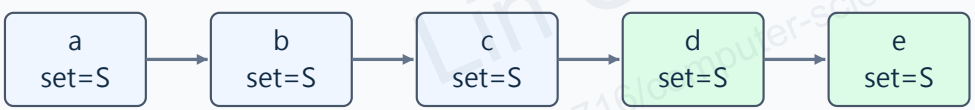
A set record stores head, tail, size. Each member stores next and a set pointer. FIND(x) returns x.set.head in O(1).

Weighted lists: update set as well as next

Before: S head=a, tail=c, size 3; T head=d, tail=e, size 2.



Append short to long. Change only d,e set pointers to S, and set c.next=d.



After: S head=a, tail=e, size 5. Green members changed set; arrows show next.

Merging a→b→c with d→e changes tail c→e and size 3→5. **Both d.set and e.set must point to the retained record** before their finds return a.

Finish every step of weighted-list UNION

Let S=x.set and T=y.set. Return for the same record; otherwise retain the longer list as S, with first-set ties.

Order	Update	Purpose
1	S.tail.next=T.head	Join the chains
2	Preserve T's old head; update S.tail and S.size	Keep the traversal start and update metadata
3	Follow T's next links; redirect every member's set to S	Make their finds return S.head
4	Discard T's record only after all redirects	Leave no dangling set pointers

Do not change T's internal next links during traversal. The merge costs $\Theta(1+\min(|S|,|T|))$: **a tail pointer makes joining constant-time, not the whole UNION.**

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Why can unweighted lists take quadratic work?

Always append the second set to the first. In UNION(x_2,x_1), UNION(x_3,x_2), …, each new singleton absorbs the existing long list.

Merge	Members whose set pointers change
UNION(x_2,x_1)	1
UNION(x_3,x_2)	2
…	…
UNION(x_n,x_{n-1})	n−1

Total rewrites are n(n−1)/2. There are only n creations and n−1 unions, so m=2n−1, yet the cost is $\Theta(n^2)$.

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Moving the shorter list: how often can one member move?

Fix member x. Whenever its set pointer changes, it belonged to the shorter list; its new set is **at least twice as large**.

After k moves its set has at least 2^k members, at most n. Thus $k \leq \lfloor \log_2 n \rfloor$. Sum over members and add the m constant call overheads: **$O(m+n \log n)$**, or $O(m+n \log(n+1))$ including small n.

Quantity	Bound or example
Whole sequence	$O(m+n \log n)$
One merge of two n/2-member sets	Still $\Theta(n)$
Balanced rounds with $n=2^k$	n/2 rewrites per round; $(n/2)\log_2 n$ total

> A sequence bound is not a per-call worst-case bound. Weighted union does not make every sequence linear.

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Eight elements: exactly which members move?

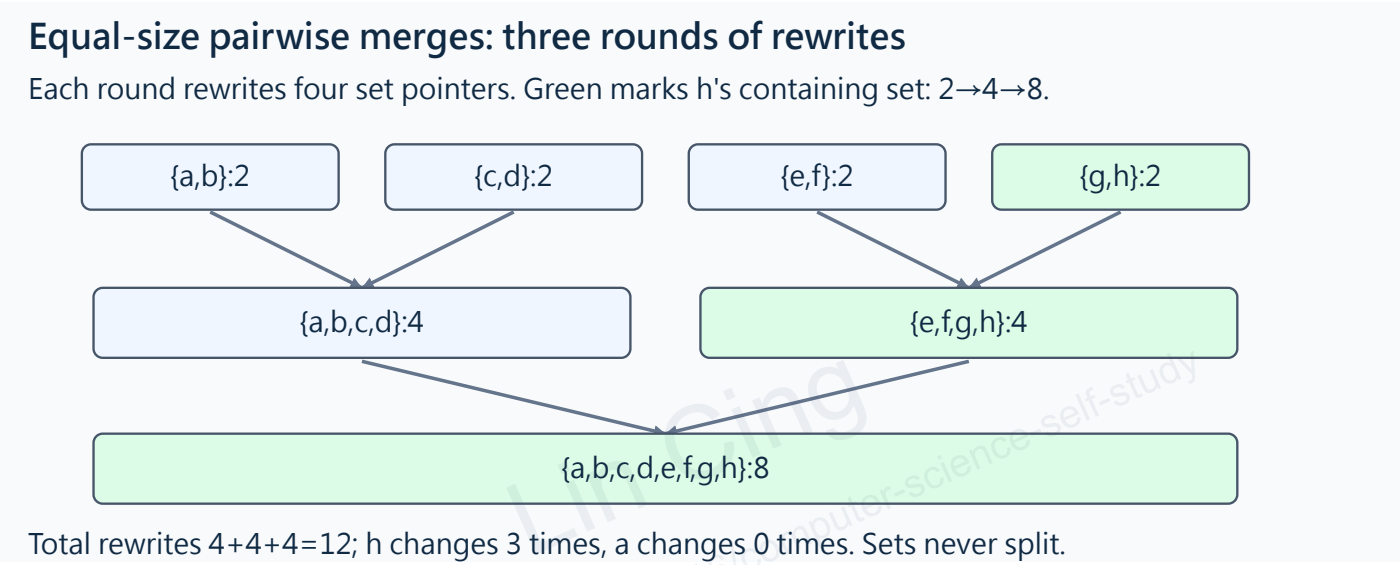
Retain the first set on equal sizes. The same operations will later build a forest.

UNION	Old sizes	Members rewritten	New representative / size	Cumulative
a,b	1,1	b	a / 2	1
c,d	1,1	d	c / 2	2
e,f	1,1	f	e / 2	3
g,h	1,1	h	g / 2	4
a,c	2,2	c,d	a / 4	6
e,g	2,2	g,h	e / 4	8
a,e	4,4	e,f,g,h	a / 8	12

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Member h follows h→g→e→a as its set grows 1→2→4→8; a never moves. **Twelve total rewrites do not mean every member moves three times.**

One diagram connects round totals to one member's growth



Round totals and one member's move count have different denominators. The table sums to 12 rewrites; h alone moves three times.

Attaching only new singletons to a large set can instead take n−1 rewrites. This meets the upper bound; it does not make every possible sequence linear.

Forests: find the roots before linking

Each element stores parent and rank; a root points to itself. MAKE-SET sets parent[x]=x and rank[x]=0. UNION must first find both roots. Linking arbitrary x directly to y can split an existing set or create a cycle.

```
UNION(x,y):
    r=FIND(x); s=FIND(y)
    if r==s: return r
    return LINK(r,s)
LINK(r,s):                    // r and s are distinct roots
    if rank[r]<rank[s]: swap r,s
    parent[s]=r
    if rank[r]==rank[s]: rank[r]+=1
    return r
```

After linking roots, all old members still reach the new root. One root loses its self-parent, with no reverse path creating a cycle. Unequal ranks leave the retained rank unchanged; equal ranks increase it by one. Rank forests and size-weighted lists are different representations. Rank is not exact set size.

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Two-pass FIND: locate the root, then follow the old path

Starting at x, follow parent pointers until reaching a self-parent root. Pass one only finds root r. Pass two redirects each node on the **original search path** directly to r.

```
FIND(x):
    r=x
    while parent[r]!=r: r=parent[r]
    while parent[x]!=x:
        old=parent[x]
        parent[x]=r
        x=old
    return r
```

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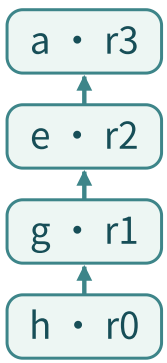
Variable old preserves the next stop before mutation. Using x=parent[x] instead reads the root just written and skips intermediate nodes. Recursive parent[x]=FIND(parent[x]) redirects nodes as calls return.

The two-pass version uses O(1) auxiliary space; the recursive stack scales with original path height. Both take Θ(s) work for s visited nodes, change only this path, and preserve ranks.

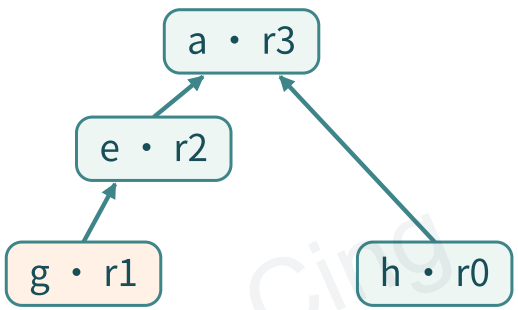
Finding the right root is not full path compression

Arrows point **child→parent**. Only h→g→e→a is drawn; a is self-parent.

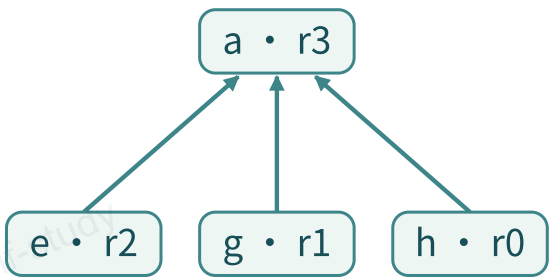
Original path



Jump to root after writing



Save old parent; finish path



Current node in pass two	Save old	Assign parent	Continue at
h	g	a	g
g	e	a	e
e	a	a, unchanged	a, then stop

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The same eight elements before and after compression

For the seven earlier unions, equal ranks retain the first root. The successive root ranks are a1,c1,e1,g1,a2,e2,a3. Final parents are b→a, c→a, d→c, e→a, f→e, g→e, h→g; a points to itself.

Vertex	a	b	c	d	e	f	g	h
rank	3	0	1	0	2	0	1	0
Parent before FIND(h)	a	a	a	c	a	e	e	g
Parent after FIND(h)	a	a	a	c	a	e	a	a

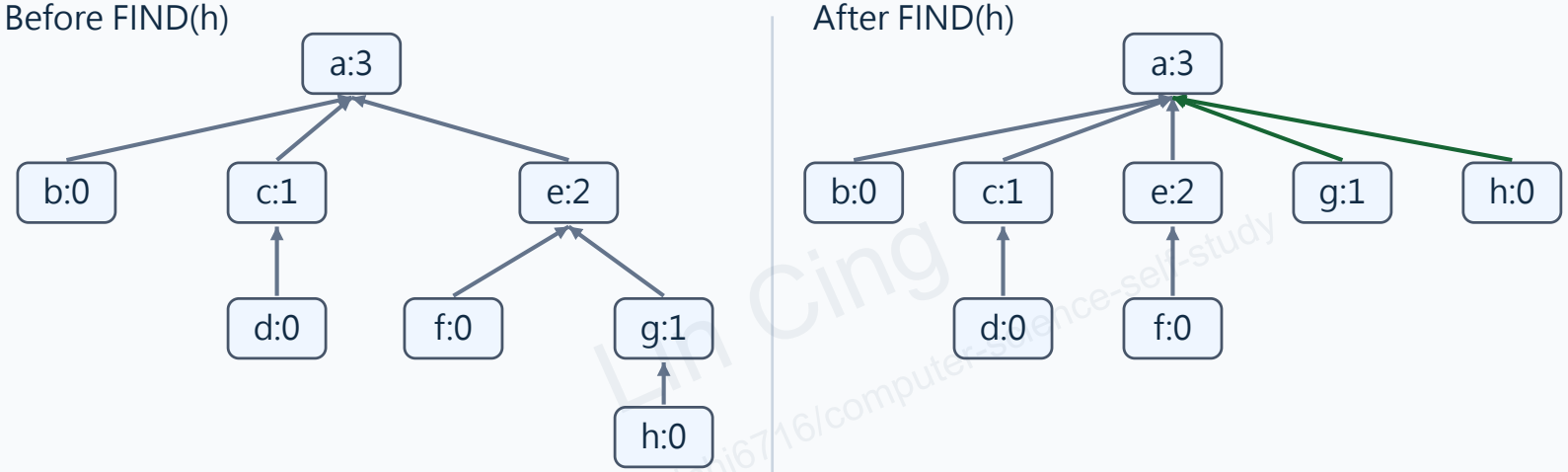
- Pass one follows h→g→e→a: three parent edges.
- Only **h and g change parent**; e already points to a. Off-path nodes do not change.
- **Every rank stays fixed.** Compression does not replace historical ranks with current heights.

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Read the tree: redirect the path, retain the branches

The same eight-node tree: parents before and after compression

Nodes show key:rank; arrows show parent; root self-loops omitted; green links changed.



h→g→e→a; only h,g links change. Node e already points to a; d,f are off-path.

A later FIND(h) takes one edge; FIND(d) still follows d→c→a. **One FIND compresses only its own search path.**

Why does rank bound tree height?

Rank is an **upper bound**, rather than a measurement of current height. Induct on operations:

Operation	Effect on height	Why height $\leq$ rank survives
Make a singleton	Height is 0	Initial rank is also 0
Merge unequal ranks	Attach the lower-rank root	The rank gap pays for its extra parent edge
Merge equal ranks	Height increases by at most 1	The retained root's rank also increases by 1
Compress a path	Redirect to ancestors, shortening paths	Ranks stay fixed

> **Do not recompute rank after compression.** A rank-2 root may now have height 1.

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Rank remembers the members accumulated during merging

A root's rank rises only by merging two different sets of equal rank:

- Rank 0 requires at least one member.
- Two rank-(r−1) sets each contain at least 2^{r-1} members. Their merged rank-r root has at least 2^r.
- Unequal-rank merges add members; FIND preserves set membership.

A nonroot's rank freezes when it loses root status. **Its set then had at least 2^r members.** Thus $2^r \leq n$, giving $r \leq \lfloor \log_2 n \rfloor$. This is not a bound on its current subtree size.

> Rank alone gives $O(\log n)$ per FIND. Compression improves the **amortized sequence bound**; a first deep-leaf query can still traverse $\Theta(\log n)$ edges.

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21.4 Fine-grained analysis of rank and compression

Ackermann iteration means composition

Fix the final number of elements n and set $\alpha = \alpha(n)$ throughout the analysis, even while MAKE-SET calls are still creating elements. For $j \geq 1$, define

$$f^{(0)}(j) = j, \quad f^{(i+1)}(j) = f(f^{(i)}(j)), \quad A_0(j) = j + 1, \quad A_k(j) = A_{k-1}^{(j+1)}(j) \ (k \geq 1).$$

The superscript counts **repeated applications to the previous output**. Since $A_0^{(i)}(j) = j + i$, we get $A_1(j) = 2j + 1$. Induction gives $A_1^{(i)}(j) = 2^i(j + 1) - 1$, hence

$$A_2(j) = 2^{j+1}(j + 1) - 1, \qquad A_3(1) = A_2(A_2(1)) = A_2(7) = 2047.$$

Inductively, each A_k is strictly increasing in its input and satisfies $A_k(j) \geq j + 1$: a larger starting input undergoes more iterations, each increasing its input. Since $j + 1 \geq 2$, also $A_k(j) > A_{k-1}(j)$. Both monotonicities are used below.

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The chosen inverse-Ackermann convention

Use the single-parameter convention of CLRS third edition, Section 21.4:

$$\alpha(n) = \min\{k \geq 0 : A_k(1) \geq n\}.$$

k	0	1	2	3	4
$A_k(1)$	2	3	7	2047	greater than 10^{80}

Thus $\alpha(2) = 0$, $\alpha(8) = 3$, and $\alpha(2048) = 4$. Other texts may use a two-parameter convention; its numerical values cannot be substituted here.

The complete bound is $O(m(1 + \alpha(n)))$ for m external operations, including n MAKE-SET calls. Only for $n \geq 3$ can it be shortened to $O(m\alpha(n))$. Small instances still require constant work per operation.

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Which quantities can change?

Induction on operations shows that a root points to itself and every nonroot has smaller rank than its parent. Linking equal-rank roots raises the retained root's rank by one. Unequal ranks attach the smaller rank under the larger. Compression changes parents, never ranks.

Once a node becomes a nonroot, its rank is fixed. Its parent's rank may increase, or compression may replace the parent by a higher ancestor. Neither event decreases parent rank. This is why a node's level cannot decrease over time.

A rank-r root contains at least 2^r elements. A nonroot retains that historical bound from losing root status. Hence every rank is at most $\lfloor \log_2 n \rfloor < n$, regardless of the current compressed subtree size.

Define level and iter only for positive-rank nonroots. Rank-zero nodes and roots receive separate potential formulas.

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Level selects the scale; iter measures progress within it

For a nonroot, write $r = \operatorname{rank}(x) \geq 1$, $p = \operatorname{rank}(\operatorname{parent}(x))$, and define

$$\ell(x) = \max\{k \geq 0 : p \geq A_k(r)\}, \qquad i(x) = \max\{j \geq 1 : p \geq A^{(j)}_{\ell(x)}(r)\}.$$

Here $\ell = \operatorname{level}$ and $i = \operatorname{iter}$. First find the highest scale reached by parent rank; then count iterations at that scale.

Since $p \geq r + 1 = A_0(r)$, we have $\ell \geq 0$. Also $A_\alpha(r) \geq A_\alpha(1) \geq n > p$, so $\ell < \alpha$. The definition gives $i \geq 1$. If $i \geq r + 1$, then

$$p \geq A^{(r+1)}_\ell(r) = A_{\ell+1}(r),$$

contradicting maximality of ℓ. Therefore $0 \leq \ell < \alpha$ **and** $1 \leq i \leq r$.

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Nonnegative potential records work still available

Assign a potential to each node and sum over the forest:

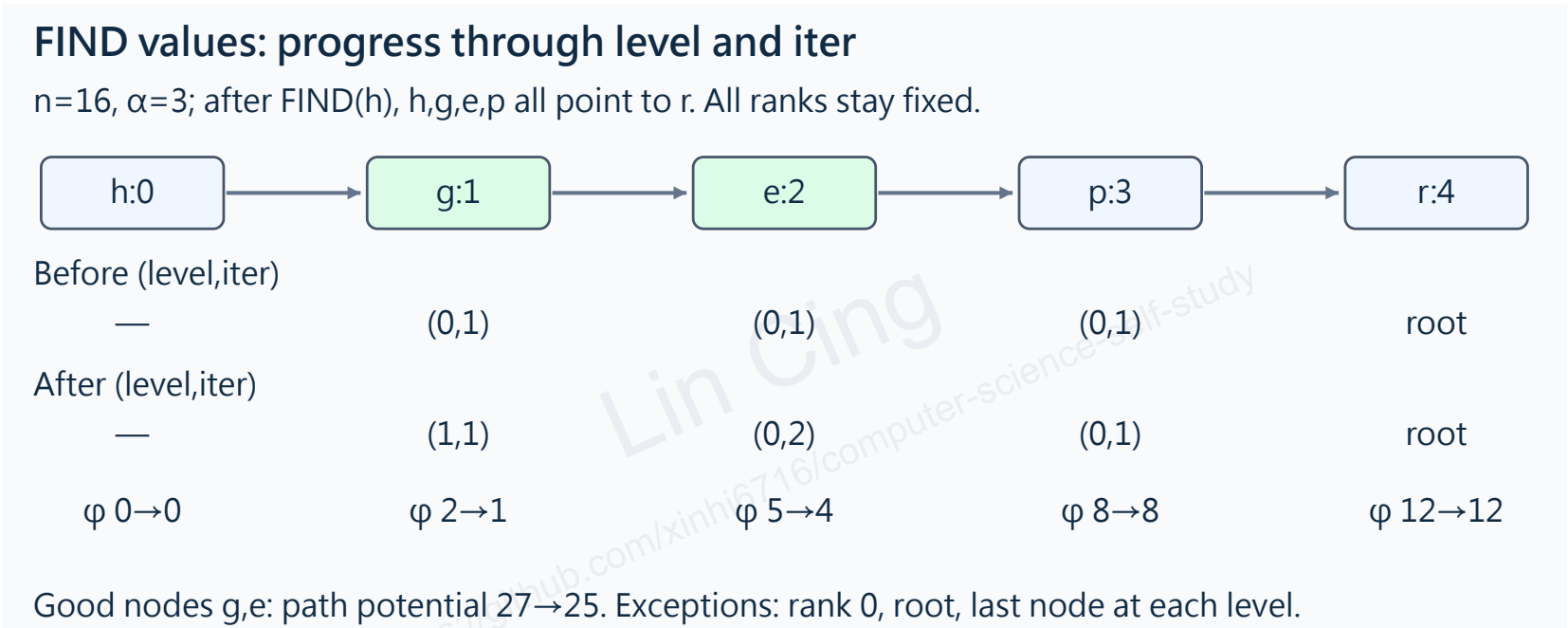
$$\phi(x) = \begin{cases} \alpha r, & x \text{ is a root or } r = 0, \\ (\alpha - \ell(x))r - i(x), & x \text{ is a nonroot and } r \geq 1, \end{cases} \qquad \Phi = \sum_x \phi(x).$$

For nonroots, the lower bound is $(\alpha - (\alpha - 1))r - r = 0$, and the upper bound is $\alpha r - 1$. Roots have potential αr; rank-zero nodes have zero potential. Thus Φ is always nonnegative.

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Potential accounting on a five-node path

Take $n = 16$, $\alpha = 3$, and compress a path of ranks 0, 1, 2, 3, 4. The root keeps its self-parent.



Each good node releases one unit: g gains a level; e increases iter within one level. The other three potentials stay fixed, so the path total falls from 27 to 25.

Why can a rising level reset iter?

Fix the rank r of a nonroot. Within one level, increasing parent rank leaves iter unchanged or increases it. Each additional iteration decreases ϕ by one.

If the level rises from k to $k' > k$, iter may reset from i to a smaller i'. Nevertheless,

$$\Delta\phi = -(k' - k)r + (i - i') \leq -r + (r - 1) = -1.$$

This uses $k' - k \geq 1$, $i \leq r$, and $i' \geq 1$. **The gain from changing level pays for resetting iter.** The pair progresses in lexicographic order, comparing level first, then iter. Its two coordinates need not increase independently.

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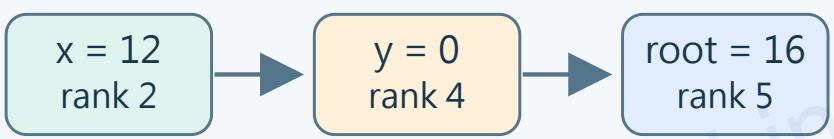
A legal FIND that really decreases iter

Take $n = 32$ and $\alpha = 3$. The find path is $12 \to 0 \to 16$, with ranks $2, 4, 5$. Unvisited branches are omitted and remain unchanged.

A higher level can reset iter from 2 to 1

n=32, α=3; FIND(12)

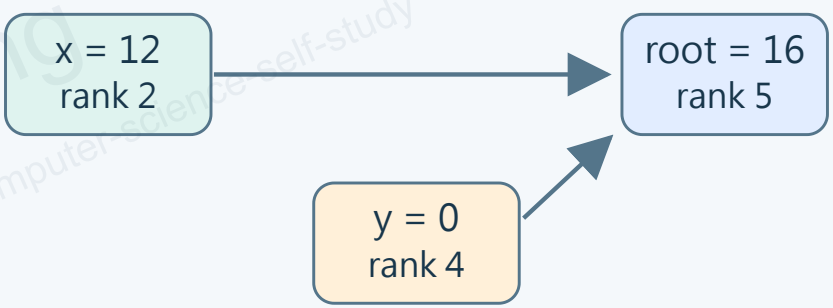
Before



x: (level, iter, φ) = (0, 2, 4)

y: φ = 11; root: φ = 15

After



x: (level, iter, φ) = (1, 1, 3)

Only the find path is drawn; omitted branches are unchanged.

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Computing the reset example term by term

Node $x = 12$ keeps rank 2. Only its parent rank changes, from 4 to 5.

Quantity	Before compression	After compression
level	$A_0(2) = 3 \le 4 < A_1(2) = 5$, so 0	$A_1(2) = 5 \le 5 < A_2(2) = 23$, so 1
iter	$A_0^{(2)}(2) = 4 \le 4 < A_0^{(3)}(2) = 5$, so 2	$A_1(2) = 5 \le 5 < A_1^{(2)}(2) = 11$, so 1
$\phi(x)$	$(3-0)2 - 2 = 4$	$(3-1)2 - 1 = 3$

Node 0 keeps potential 11, and the root keeps potential 15. All other branches are unchanged. Thus the forest has $\Delta\Phi = (3 + 11 + 15) - (4 + 11 + 15) = -1$. The displayed totals sum only the path; the other 29 nodes need not have zero potential.

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Constructing that path with legal operations

Create singleton keys 0 through 31. Equal-rank merges retain the first root.

Operation	Checkable result
Merge adjacent equal-size blocks of 0..15 in rounds	Root 0 has rank 4; node 12 has rank 2
FIND(12)	Parent of 12 becomes 0; ranks stay fixed
Build 16..31 by the same rounds	Root 16 has rank 4
UNION(16,0)	Root 16 rises to rank 5; parent of 0 becomes 16
FIND(12) again	Path $12 \to 0 \to 16$; parent of 12 becomes 16

The first half starts with (0,1), (2,3), …, (14,15), then (0,2), (4,6), …, followed by (0,4), (8,12), and finally (0,8). These ranks come from a legal history rather than arbitrary assignments to a drawn chain.

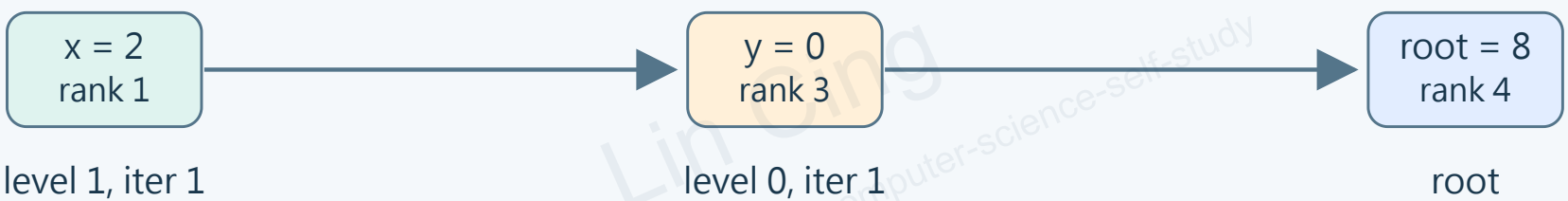
Lin Cing https://github.com/xinhi6716/computer-science-self-study

Levels need not increase along a path

Take $n = 16$. Merge 0..7 into root 0 of rank 3, then call FIND(2). Build 8..15 into root 8 of rank 3 and call UNION(8,0). This creates the following legal path.

Increasing ranks do not imply increasing levels

n=16, α=3; FIND(2)



$A_1(1)=3 \le \text{rank}(y) < A_2(1)=7$; $A_0(3)=4 \le \text{rank}(\text{root}) < A_1(3)=7$

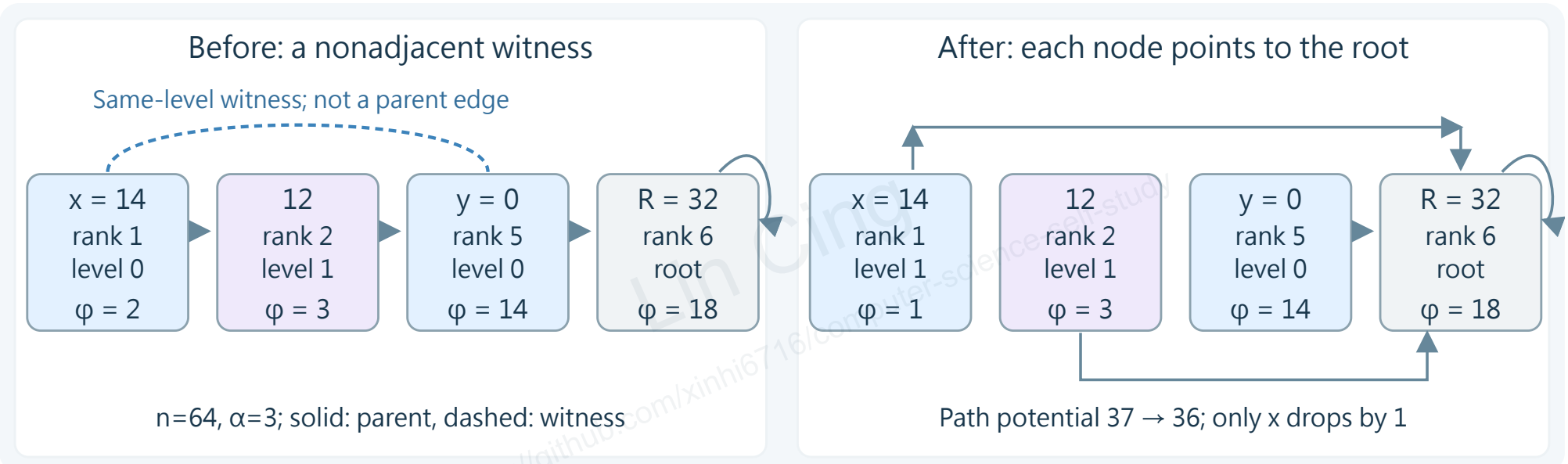
Along this path: ranks 1 < 3 < 4, but levels 1 > 0.

Count the last node of each level; no sorted-level assumption is needed.

Comparing different nodes changes the input rank too. Temporal monotonicity concerns one fixed node.

A same-level witness need not be the next node

A positive-rank nonroot x is good if a later nonroot y on the old path has the same level. **The witness need not be x's parent.**



In 14→12→0→32, levels are 0,1,0. Node 0 witnesses that 14 is good; node 12 may have another level. The root has no level.

Construct the path through legal operations

Start with 64 singletons. Equal-rank UNION retains its first root. Building an interval in rounds means merging adjacent singletons, then adjacent equal-sized blocks.

Operation	Fact established by the history
Build 0..31 in rounds	Root 0 has rank 5; node 12 has rank 2 and node 14 rank 1
FIND(12)	Old path is $12 \to 8 \to 0$; attach 12 to 0, while 14 still points to 12
Build 32..63 in rounds	Root 32 has rank 5
UNION(32,0)	Equal ranks: 32 rises to 6, and 0 points to 32
Before FIND(14)	Path $14 \to 12 \to 0 \to 32$, with ranks 1,2,5,6

FIND(12) follows $12 \to 8 \to 0$ upward to the root, then attaches each visited nonroot directly to that root. It never enters a child subtree, so 14 still points to 12. The later union attaches root 0 to root 32. These ranks follow from legal operations; 32 retains its self-parent.

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Compute every level, iteration count, and potential

Fix $n = 64, \alpha = 3$. Level and iter apply only to positive-rank nonroots; the root uses $\phi = \alpha r$.

Node	Rank	Parent rank: old→new	Level: old→new	Iter: old→new	ϕ: old→new
14	1	2→6	0→1	1→1	2→1
12	2	5→6	1→1	1→1	3→3
0	5	6→6	0→0	1→1	14→14
32, root	6	6→6	Not defined	Not defined	18→18

For 14, initially $A_0(1) = 2 \le 2 < A_1(1) = 3$, so level=0,iter=1. Afterwards $A_1(1) = 3 \le 6 < A_2(1) = 7$, and $A_1^{(2)}(1) = 7 > 6$, so level=1,iter=1. Its potential drops from $(3-0)1 - 1 = 2$ to $(3-1)1 - 1 = 1$.

For 12, $5 = A_1(2) \le p < 11 = A_1^{(2)}(2) < A_2(2) = 23$, so level=1,iter=1. For 0, $A_0(5) = 6 \le 6 < A_0^{(2)}(5) = 7 < A_1(5) = 11$, so level=0,iter=1,potential=14. Path total changes $37 \to 36$. All other nodes keep rank,parent rank,root status; **the whole forest also loses exactly one unit**.

The good-node inequality, with a reason for each step

Freeze all quantities before compression. Write $\rho(v) = \operatorname{rank}(v)$, $p_x = \operatorname{parent}(x)$, $p_y = \operatorname{parent}(y)$, and let R be the root. Put $r = \rho(x), k = \ell(x) = \ell(y), i = i(x)$.

Step	Reason
$\rho(R) \geq \rho(p_y)$	p_y is on the root path; rank strictly increases along nonroot parent edges
$\rho(p_y) \geq A_k(\rho(y))$	Definition of $\ell(y) = k$
$A_k(\rho(y)) \geq A_k(\rho(p_x))$	y equals or is above p_x, so $\rho(y) \geq \rho(p_x)$; A_k increases
$A_k(\rho(p_x)) \geq A_k(A_k^{(i)}(r))$	Iter gives $\rho(p_x) \geq A_k^{(i)}(r)$; use monotonicity again
$A_k(A_k^{(i)}(r)) = A_k^{(i+1)}(r)$	One more function composition, not multiplication

The new parent rank is therefore at least $A_k^{(i+1)}(r)$. No step assumes sorted path levels: the proof uses ancestor rank order and monotonicity at one fixed level k.

Crossing a threshold pays for a good node

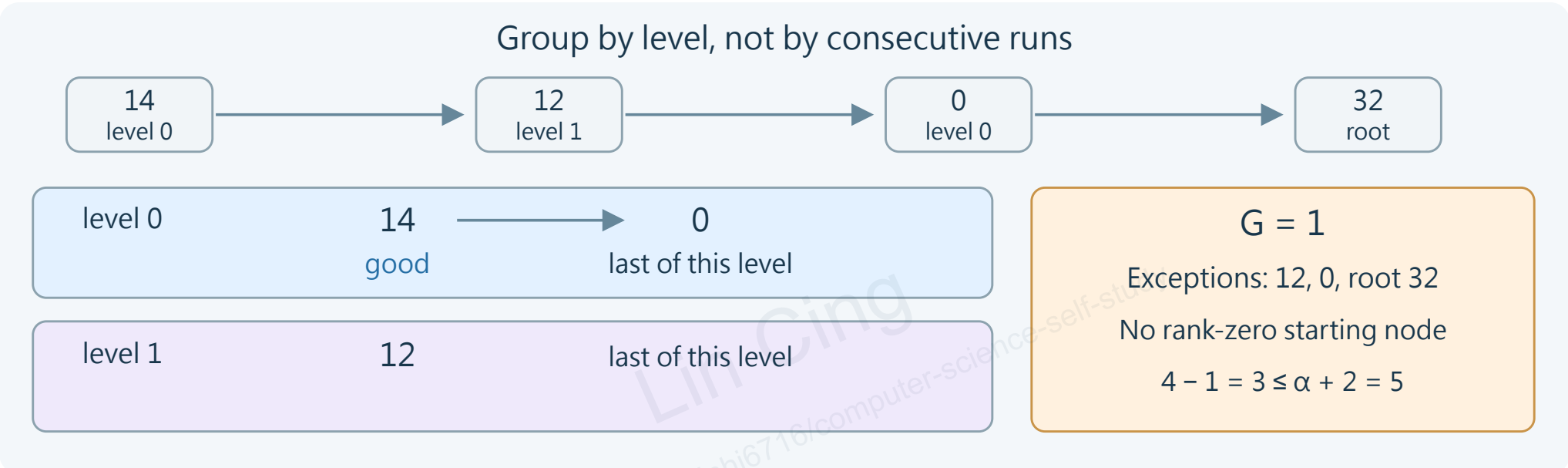
Compression makes R the parent of x, keeping r fixed. The proof crosses the **next iteration threshold of the old level**. It does not guarantee that the new iter exceeds the old one; distinguish two cases.

After compression	Potential change	Why it decreases
Level stays k	$\Delta\phi = -(i' - i) \le -1$	Iter is at least $i + 1$
Level rises	$\Delta\phi = -(k' - k)r + i - i'$	$k' - k \ge 1, i \le r, i' \ge 1$, so $\Delta\phi \le -r + (r - 1) = -1$

In the 64-node example, $x = 14, y = 0, k = 0, i = 1$; the inequalities become $6 \ge 6 \ge 6 \ge 3 \ge 3$. The next old-level threshold is $A_0^{(2)}(1) = 3$. New parent rank 6 also crosses $A_1(1) = 3$, so this is the second case and potential drops by one.

Charge the decrease to good node x, not again to witness y; y's potential need not drop. Other nodes' potentials do not increase, so the good-node decreases may be summed.

Only the last node of each level is an exception



Within one level, every node except the last has a later same-level witness. Here 14 is good; 12 and 0 are the last nodes of levels 1 and 0. Adding root 32 gives three exceptions. Treating consecutive runs 0/1/0 as three groups would miss good node 14.

Convert group counts into total potential decrease

An s-node path has at most α positive-rank nonroot levels, numbered $0, \ldots, \alpha - 1$. At most one last node per level, the root, and a possible rank-zero starting node give at most $\alpha + 2$ exceptions. Strictly increasing ranks allow at most one rank-zero node.

$$G \geq g := \max(0, s - (\alpha + 2)), \qquad \Delta\Phi \leq -G \leq -g.$$

G counts actual good nodes; g is only a conservative lower bound. Here $s = 4, \alpha = 3$, hence $g = 0$, yet $G = 1$ and $\Delta\Phi = -1$. A short path need not attain the bound. The next argument uses the general lower bound to pay for long paths.

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Fixed constants pay for the entire scan

Suppose two-pass find costs at most cs. Fix $C > c$ and use scaled potential $\Psi = C\Phi$. The amortized cost satisfies

$$\hat{c} = c_{\text{actual}} + \Delta\Psi \le cs - Cg.$$

Path length	Bound after substituting g
$s \le \alpha + 2$	$cs \le c(\alpha + 2)$
$s > \alpha + 2$	$(c - C)s + C(\alpha + 2) \le C(\alpha + 2)$

Thus each FIND has amortized cost $O(1 + \alpha)$. For $s = 7$, $\alpha = 3$, $c = 2$, $C = 3$, the lower bound is $g = 2$, so $\hat{c} \le 14 - 6 = 8$. Keep the same C throughout the sequence.

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Only the retained LINK root can gain potential

MAKE-SET adds a rank-zero node with zero potential and still has amortized cost $O(1)$. Suppose LINK attaches old root u under retained root v.

Possibly affected nodes	Reason for the potential change
u	Loses root status without changing rank; potential does not rise
Old children of v	Parent rank cannot fall; the earlier monotonicity argument applies
v	Unequal ranks leave it fixed; equal ranks add one, increasing potential by α

Every other node keeps its rank, parent rank and root status. Therefore $\Delta\Psi \le C\alpha$, and LINK costs $O(1) + C\alpha = O(1+\alpha)$ amortized. UNION performs two FIND calls and links only distinct roots, expanding into at most three operations.

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Recovering actual sequence cost

Let c_j be actual cost and $a_j = c_j + \Psi_j - \Psi_{j-1}$ amortized cost. Summing cancels intermediate potentials:

$$\sum_{j=1}^{m'} c_j = \sum_{j=1}^{m'} a_j + \Psi_0 - \Psi_{m'} \leq \sum_{j=1}^{m'} a_j.$$

The inequality uses the empty initial forest, $\Psi_0 = 0$, and nonnegative final potential. Expanding UNION gives $m' \leq 3m$, so the original m operations cost $O(m(1 + \alpha(n)))$ in total.

This is a worst-case bound on the entire sequence. With rank and compression, a single FIND can still cost $\Theta(\log n)$: build a balanced rank tree and first query a deepest leaf. A linear chain requires removing the rank rule and does not satisfy this theorem's hypotheses.

Potential analysis reference: CLRS, third edition, Section 21.4, printed pp. 576–581.

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References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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